

Module 1: Functions – Prequiz

Name:

Date:

Four short questions. Show enough work that you can find the step where an error occurred.

1. Consider the relation

$$R = \{(-2, 5), (0, 1), (3, 4), (0, -2)\}.$$

Does this relation define a function? Explain briefly.

2. Let $f(x) = 2x^2 - 3x + 1$. Find $f(-2)$.

3. The graph of $y = f(x)$ is transformed to

$$g(x) = f(x - 3) - 2.$$

Describe the horizontal and vertical shifts in words.

4. The point $(1, 4)$ lies on the graph of $y = f(x)$. Where does this point move on the graph of

$$y = -2f(x + 1) + 3?$$

Module 1: Functions – Weekly Check-In

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Three questions. Show enough work to make your reasoning clear.

1. Relations and functions.

Consider the relation

$$R = \{(-4, 1), (-1, 3), (2, 5), (4, 3), (-1, -2)\}.$$

Does this relation define a function? State *yes* or *no* and explain using the input-output definition of a function.

2. Function values and transformed points.

Let $f(x) = 2x^2 - 3x + 1$.

(a) Find $f(-2)$.

(b) Suppose the point $(1, 4)$ lies on the graph of $y = h(x)$. Where does that point move on the graph of

$$y = -2h(x + 1) + 3?$$

Show how the coordinates change.

Module 1: Functions – Weekly Check-In, continued

Name:

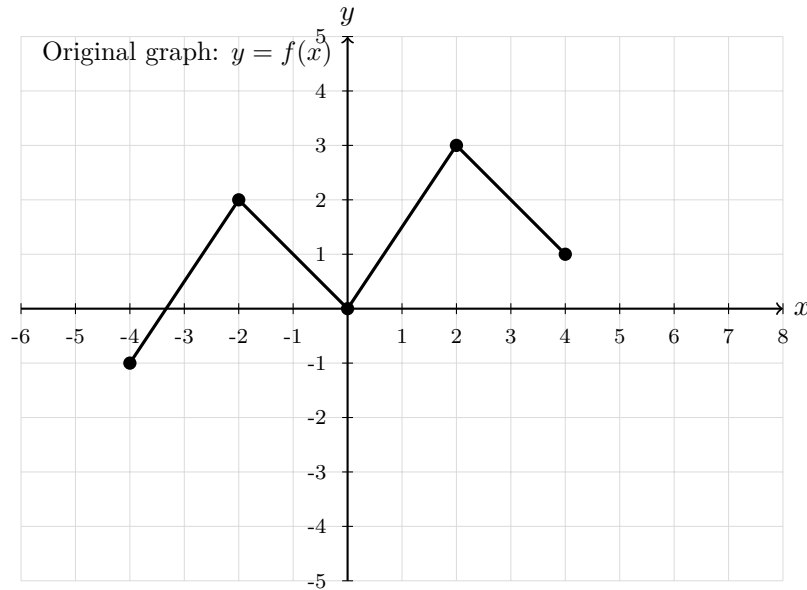
Date:

3. Transforming an unknown graph.

The graph of $y = f(x)$ is shown below. On the *same coordinate plane*, graph

$$y = -f(x - 2) + 1.$$

Show the transformed locations of all five marked points and connect them with the same shape as the original graph.



Below the graph, list the five transformed points as ordered pairs.